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Global Technologies

XD12

The LED Current Sink Driver IC

12 Channels

General Description

The XD12 is a compact IC featuring 12-channel current sink control drivers optimized for LED lighting solution. The XD12 driver is used together with a XC24 controller as a companion chip. A lot of the XD12 drivers can be daisied each other to the XC24 channels and controlled through a serial interface by a XC24 controller.

The XD12 has a feedback control for the LED power efficiency, and self-protection features like as a thermal-regulation, a temperature-protection, a LED open and a short monitoring.

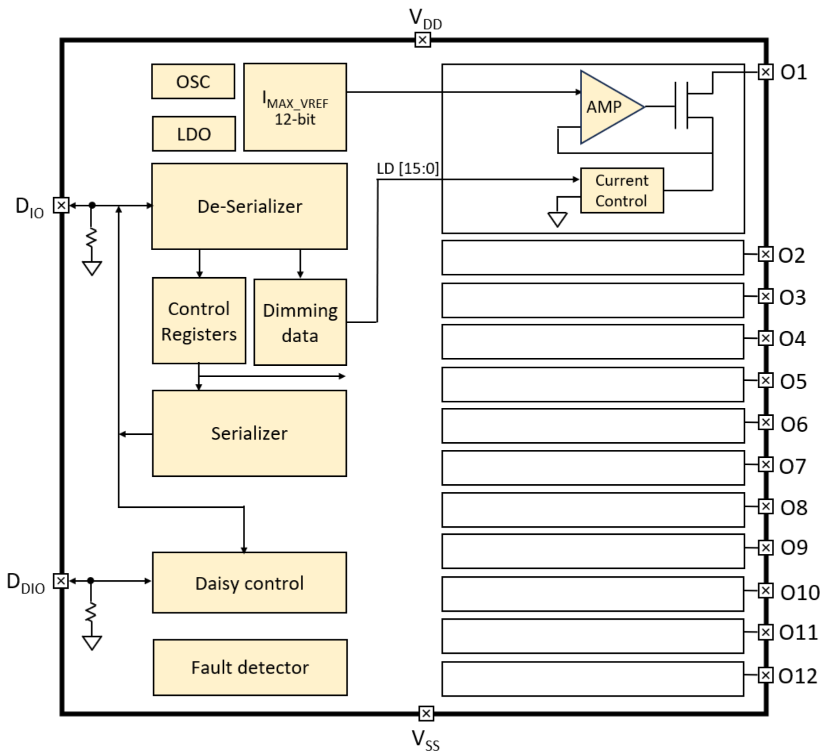
Application

LED back-lighting with local dimming control

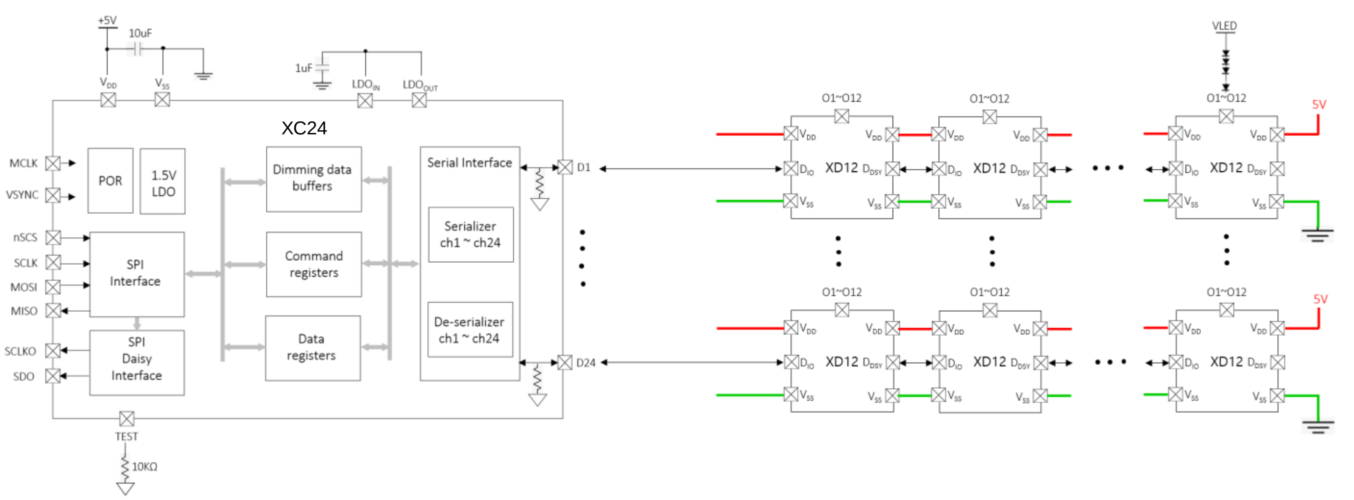
Feature

- Bi-directional 1-wire interface and configurable transmit rate
- Daisy connection and pin layout for designing single layer PCB
- 12 independent dimming control channels in a package
- Maximum 40V headroom voltage
- Configurable device's max current level (4/8/12/16/24/32/46/64mA)
- 12-bit max current control for dimming
- 16-bit brightness control (DC level) under the 12-bit max current level
- Sink current accuracy ±2%
- Sync controllability between devices
- Flicker-less frame frequency switching
- Configurable LED open detection and protection
- Configurable LED short detection level (3/4/6/8/12/24/36V) and protection
- Configurable FB operation level (0.4/0.5/0.6/0.7/0.85/1.0/1.05/1.3V)
- Configurable over-temperature protection and Auto thermal-regulation @ 140℃
- Supply voltage 5.0V
- 24-PIN QFN 3030 package
- Latch-up Immunity, I_{LATCH} (JEDEC JESD78D Nov 2011) : ±100mA
- ESD protection, ESD_{HBM} (Norm: MIL 883 E Method 3015 Human Body Model) : ±2kV

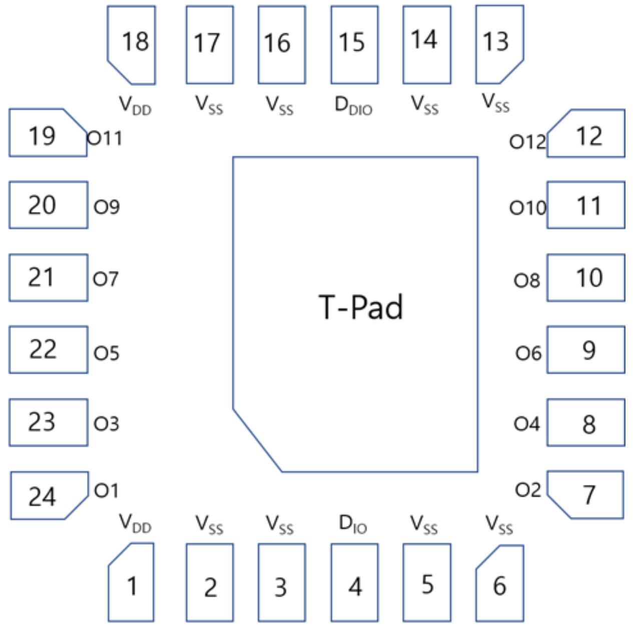
Block Diagram



Application Circuit



Pin diagram (Top View)



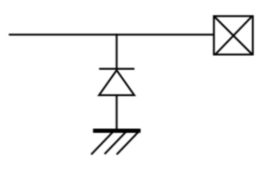
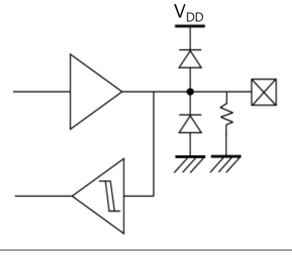
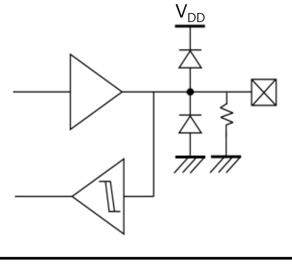
Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
V_{DD}	1	-	Power	Supply voltage	
V_{SS}	2	-	GND	Ground	
V_{SS}	3	-	GND	Ground	
D_{IO}	4	I/O	Digital	Data In/Out	Internal PD, 48KΩ
V_{SS}	5	-	GND	Ground	
V_{SS}	6	-	GND	Ground	
O2	7	O	Current Load	Current Sink 2	If unused, connect to GND
O4	8	O	Current Load	Current Sink 4	If unused, connect to GND
O6	9	O	Current Load	Current Sink 6	If unused, connect to GND
O8	10	O	Current Load	Current Sink 8	If unused, connect to GND
O10	11	O	Current Load	Current Sink 10	If unused, connect to GND
O12	12	O	Current Load	Current Sink 12	If unused, connect to GND
V_{SS}	13	-	GND	Ground	
V_{SS}	14	-	GND	Ground	

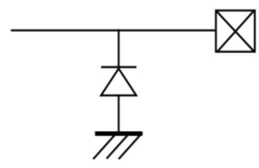
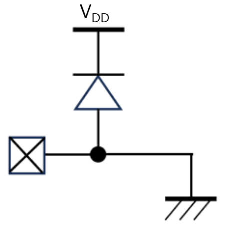
Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
D_{DIO}	15	I/O	Digital	Daisied Data In/Out	Internal PD, 48KΩ
V_{SS}	16	-	GND	Ground	
V_{SS}	17	-	GND	Ground	
V_{DD}	18	-	Power	Supply Voltage	
O11	19	O	Current Load	Current Sink 11	If unused, connect to GND
O9	20	O	Current Load	Current Sink 9	If unused, connect to GND
O7	21	O	Current Load	Current Sink 7	If unused, connect to GND
O5	22	O	Current Load	Current Sink 5	If unused, connect to GND
O3	23	O	Current Load	Current Sink 3	If unused, connect to GND
O1	24	O	Current Load	Current Sink 1	If unused, connect to GND

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
V_{DD}	-		Supply voltage
D_{IO}	I/O		Data In/Out
D_{DIO}	I/O		Daisied Data In/Out

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
O1 ~ O12	Current Sink		Current Load
VSS	-		Ground

Maximum Absolute ratings (Ta = 25℃ unless otherwise noted)

Parameter	Symbol	Value	Unit
Supply voltage	V_{DD}	-0.3 ~ 6.5	V
Sink output pin voltage	V_O	40	V
Sink output current	I_O	64	mA
D_{IO} to GND	V_{DIO}	-0.3 ~ V_{DD}+0.3	V
D_{DIO} to GND	V_{DDIO}	-0.3 ~ V_{DD}+0.3	V
Thermal Resistance			
Junction to Ambient	$R_{\theta JA}$	x [1)]	℃/W
Junction to Case	$R_{\theta JC}$	x [1)]	℃/W
Power Dissipation on package	P_{TOT}	x [1)]	W
Junction temperature	T_J	-20 ~ 150	℃
Storage temperature	T_{STG}	-45 ~ 150	℃
Relative Humidity (non-condensing)	RH_{NC}	5 ~ 85	%
Moisture Sensitivity	MSL	Maximum Floor Life-time	3

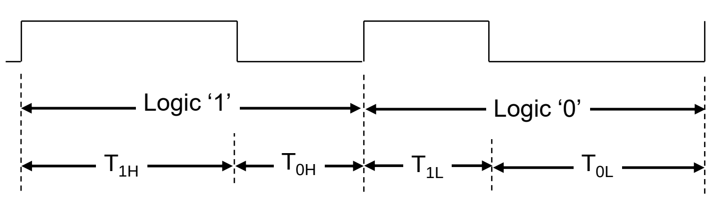
Stresses beyond the Maximum Absolute ratings may cause permanent damage to the device
Note : 1) Surface mounted on FR-4 substrate PC Board 1" x 1" inch 2-oz copper plate

Electrostatic Discharge

Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM ; Contact	ESD_{HBM}	±2K	V	C=100 pF, R=1.5 kΩ
Electrostatic Discharge CDM ; Contact	ESD_{CDM}	±500	V	R=1 Ω

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply voltage						
Operating voltage	V_{DD} 1)		4.5	5.0	5.5	V
Supply current						
Quiescent Current	I_s	Standby state, No LED current	-	-	2	mA
Operating Current 1	I_{DD1}	All output sink current = 4mA	-	-	3	mA
Operating Current 2	I_{DD2}	All output sink current = 8mA	-	-	4	mA
Operating Current 3	I_{DD3}	All output sink current = 12mA	-	-	5	mA
Operating Current 4	I_{DD4}	All output sink current = 16mA	-	-	6	mA
Operating Current 5	I_{DD5}	All output sink current = 24mA	-	-	7	mA
Operating Current 6	I_{DD6}	All output sink current = 32mA	-	-	8	mA
Operating Current 7	I_{DD7}	All output sink current = 46mA	-	-	9	mA
Operating Current 8	I_{DD8}	All output sink current = 64mA	-	-	12	mA
Oscillator						
Internal MCLK	f_{OSC}		-3%	14.746	+3%	MHz
In/Out						
D_{IO}, D_{DIO}	V_{IL}	Input logic low level	-	-	0.5	V
	V_{IH}	Input logic high level	V_{DD} -0.5	-	V_{DD}	V
	V_{OL}	Output logic low level	-	-	0.5	V
	V_{OH}	Output logic high level	V_{DD} -0.5	-	V_{DD}	V
	R_{PD}	Internal pull-down resistance	-	48	-	KΩ
Logic high time 2)	T_{1H}, T_{0L}		6	8	20	MCLK
Logic low time 2)	T_{1L}, T_{0H}		3	4	10	MCLK
Propagation delay	T_{D_DS}	Downstream (D_{IO} to D_{DIO})	-	-	2	ns
Propagation delay	T_{D_US}	Upstream (D_{DIO} to D_{IO})	-	-	2.7	ns



[Figure. 1]

Note : 1) The operation below V_{DD}, 5V will limit a maximum LED sinking current and affect a current variation.
2) $T_{1H} \geq T_{0H} * 2$, $T_{0L} \geq T_{1L} * 2$, $(T_{1H} + T_{0H}) = (T_{0L} + T_{1L}) \leq 31$

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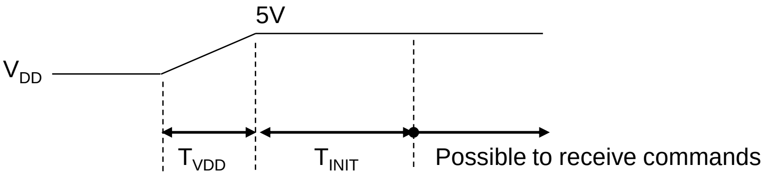
Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value min	Value typ	Value max	Unit
Output						
Driving voltage	V_O	Headroom voltage V_{HD} : 0.65V@ I_O 4mA, 0.80V@ I_O 8mA 0.90V@ I_O 12mA, 0.65V@ I_O 16mA 0.80V@ I_O 24mA, 0.90V@ I_O 32mA 0.80V@ I_O 46mA, 0.90V@ I_O 64mA	V_{HD}	-	40	V
Short detection voltage 3)	V_{SD}	Short detection level V_{SD} : 3b'000=3V, 3b'100=12V 3b'001=4V, 3b'101=16V 3b'010=6V, 3b'110=24V 3b'011=8V, 3b'111=36V	V_{SD}	-	-	V
Open detection voltage 3)	V_{OD}		-	-	0.25	V
Sink current	I_O	Device max current level : I_O 3'b000 = 4mA, 3'b001 = 8mA 3'b010 = 12mA, 3'b011 = 16mA 3'b100 = 24mA, 3'b101 = 32mA 3'b110 = 46mA, 3'b111 = 64mA	-	-	64	mA
Current accuracy 4)	%	full range @ I_O = 4mA full range @ I_O = 8mA full range @ I_O = 12mA full range @ I_O = 16mA	-2	-	+2	%
		full range @ I_O = 24mA full range @ I_O = 32mA	-3	-	+2	%
		full range @ I_O = 46mA full range @ I_O = 64mA	-4	-	+5	%
LED off current	I_{OFF}	Vo = 40V	-	-	300	nA
Feedback						
FB threshold voltage (Logically ORed for 12-ch)	V_{FBA}	FB level V_{FBA} : 3'b000: = 0.40V, 3'b001: = 0.50V 3'b010: = 0.60V, 3'b011: = 0.70V 3'b100: = 0.85V, 3'b101: = 1.00V 3'b110: = 1.00V, 3'b111: = 1.35V	-	V_{FBA}	-	V
FB hysteresis voltage	V_{FBH}		40	60	110	mV
Thermal protection						
Automatic Thermal Regulation Range @T_J	T_{ATR}		105	-	≈140	℃
Thermal shut down 3) @Tj	T_{OTP}		-	140	-	℃
Thermal shut down hysteresis @Tj	T_{HYS}		10	20	30	℃

Note : 3) If LED open or short event is happened, the corresponding channel will be offed under the function enabled.
If the thermal is happened, all channels output will be offed under the function enabled.
4) Under satisfying the power dissipation specification.

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
V_{DD} ramp up						
V_{DD} Ramp up time	T_{VDD}				1	ms
Initialization after V_{DD}, 5V stable						
Internally required Init	T_{INT}		5			ms
Allowed 1st command	T_{FCMD}		5			ms



[Figure. 2]

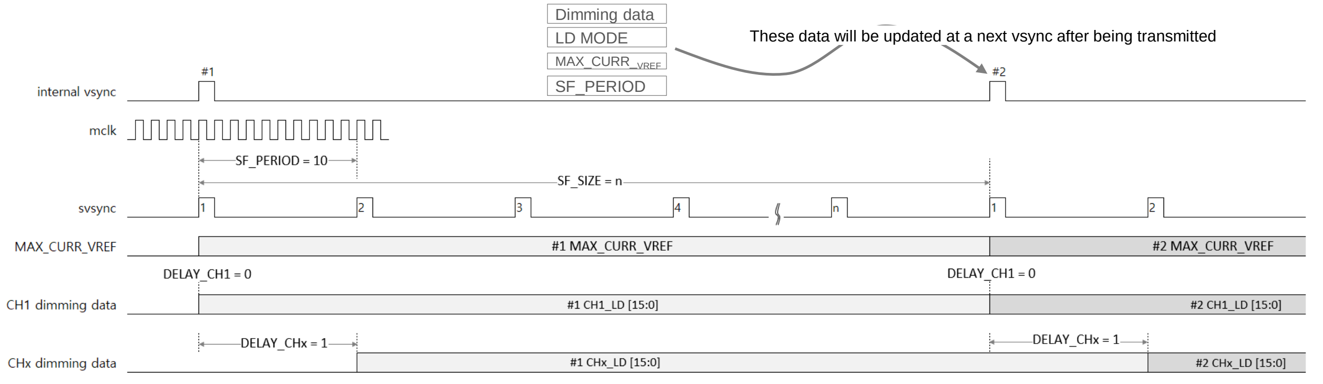
Timeout
: Internal timer starts after completing to receive a read-back command and stops after first response bit from a las daisied device

Internal Timeout 5)	$T_{TIMEOUT}$		539	556	573	us

Note : 5) If the timeout will be expired, the internal communication logic will go to an initial state, so new command transaction will be possible. The timeout function is configurable.

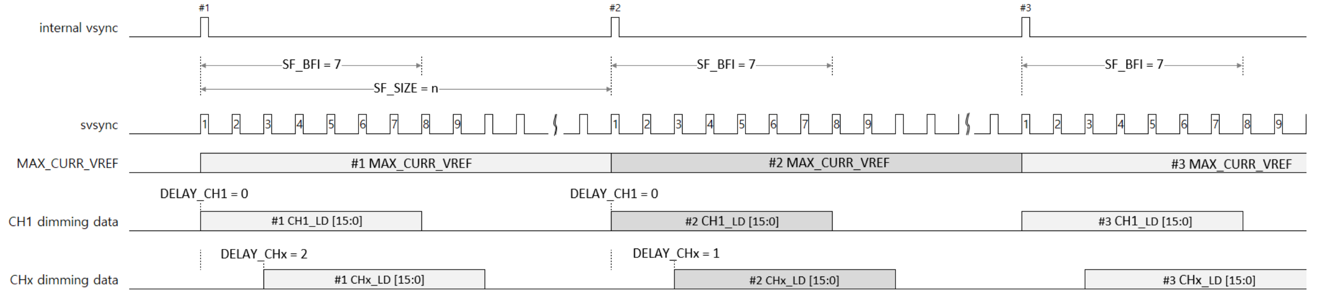
Local dimming sequence

1. Normal dimming



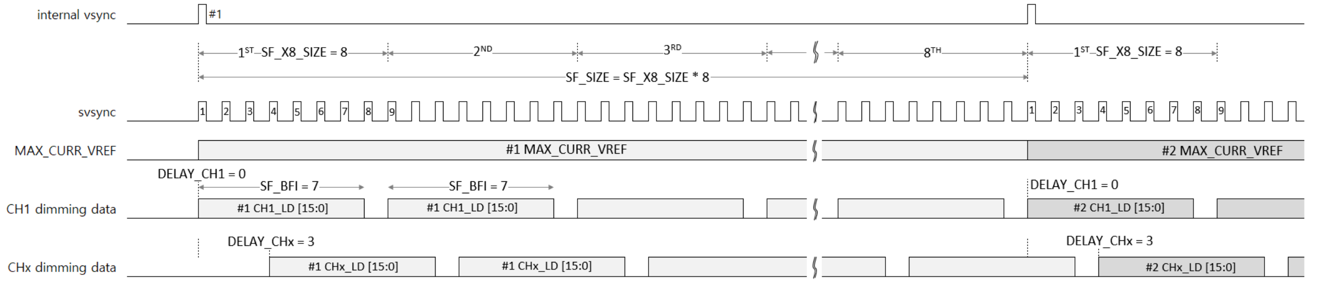
[Figure. 3] Normal dimming

2. BFI dimming



[Figure. 4] BFI dimming

3. x8 dimming



[Figure. 5] x8 dimming

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Register Map

RESET

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x0	x	x	x	x	x	x	x	x	x	x	x	RST

Bits	Fields	Descriptions
0	RST	‘1’ The internal all logics are reset and auto-cleared after setting to 1.

ID

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x0	x	x	x	x	x	x	x	ID [4:0]				

Bits	Fields	Descriptions
4:0	ID	If the XD12 will receive the ID-Gen command from the XC24 controller, then the ID value will be automatically generated. The ID value is needed for proper reading, so the ID-Gen command must be firstly executed by the XC24 controller before executing any commands. If all XD12 drivers daisied will receive the ID-Gen command, the last-located device will have the first number of ID, and the first located device will have the last number of ID, that is, the number of daisied devices.

LD MODE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x0	x	x	x	BFI	x	x	x	X8	x	x	x	NORMAL

Bits	Fields	Descriptions
		This register determines the dimming sequence. Even if the LD MODE value will be changed in the middle of frame, the dimming sequence will keep the current mode until completing current frame.
8	BFI	This mode is for inserting the black sub-frame in one frame. Refer to Figure.4.
4	X8	This mode is for 8 times repeated dimming sequence in one frame. Refer to Figure.5.
0	NORMAL	This mode is for general dimming sequence. Refer to Figure.3.

SF PERIOD

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0	SF_PERIOD [11:0]											

Bits	Fields	Descriptions
11:0	SF_PERIOD	This register determines the length of sub-frame, and if the value will be changed in the middle of a frame, the value will be applied in the next frame. For more details, refer to the Fig. 3. The sub-frame time can be calculated as follows : = SF_PERIOD * MCLK

SF SIZE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x04	0x0	SF_X8_SIZE [3:0]				SF_SIZE [7:0]							

Bits	Fields	Descriptions
11:8	SF_X8_SIZE	The SF_X8_SIZE determines the number of sub-frame to be repeated 8 times in a frame, so the set value must be divided the SF_SIZE by 8. For more details, refer to the Fig. 5.
7:0	SF_SIZE	The SF_SIZE determines the total number of sub-frames in a frame. For more details, refer to the Fig. 3/4/5.

SF BFI

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x05	0x0	SF_X8_SIZE [4]	x	x	x	SF_BFI [7:0]							

Bits	Fields	Descriptions
15	SF_X8_SIZE	This is MSB one bit of the SF_X8_SIZE register.
7:0	SF_BFI	The SF_BFI determines how many sub-frames are turn-off in a frame. This register can be activated in a X8 and a BFI mode except for the NORMAL mode. For more details, refer to the Fig. 3/4/5. The number of sub-frame turned off is as follows : = SF_SIZE – SF_BFI at a BFI mode = SF_X8_SIZE – SF_BFI at a X8 mode

LD FIX 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x0	LD_FIX [11:0]											

LD FIX 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0	x	x	x	x	x	x	x	x	LD_FIX [15:12]			

Bits	Fields	Descriptions
15:0	LD_FIX	The LD_FIX register is used to successively output a fixed value written in this register instead of LD data to be transmitted form a Host, regardless of the frame sync signal. This function is activated when the TEST_EN bit is set.

MAX CURRENT VREF

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x8	MAX_CURR_VREF [11:0]											

Bits	Fields	Descriptions
11:0	MAX_CURR_VREF	The MAX_CURR_VREF determines the maximum current value of a frame to be dimmed. The set value will come into effect at the next internal vsync assertion even if the value is set during current dimming frame earlier than the next vsync.

CHANNEL ENABLE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x0	CH12_EN	CH11_EN	CH10_EN	CH9_EN	CH8_EN	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN

Bits	Fields	Descriptions
11:0	CHx_EN	The CHx_EN determines which channel will be activated. If 9 channels are used in the device, then you must set successively from CH1_EN bit to CH9_EN bit.

FAULT STATUS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
RO	0x0B	0x0	x	x	x	x	x	x	x	x	THERMAL	SHORT	OPEN	FB

Bits	Fields	Descriptions
3	THERMAL	The THERMAL bit indicates whether the XD12 device has internally caused a thermal regulation event due to overheating due to high current for a long time. After this event has happened, this bit will be reset if internal thermal event is cleared again. This bit is set when a thermal event occurs in even one of the activated channels.
2	SHORT	The SHORT bit indicated whether the XD12 output pin is short to a VLED line due to any soldering condition. The detection level can be set by the SHORT_LEVEL of the FAULT LEVEL register and the activating level can be set by the SHORT ENABLE LEVEL register. After this event has happened, this bit will be reset if the event condition is cleared again. This bit is set when a short event occurs in even one of the activated channels.
1	OPEN	The OPEN bit indicated whether the XD12 output pin is open, even though the channel output is activated. The open level is fixed to 0.25V. After this event has happened, this bit will be reset if the event condition is cleared again. This bit is set when an open event occurs in even one of the activated channels.
0	FB	The FB bit indicated whether the XD12 output pin's headroom voltage is under the value set by FB_LEVEL of the FAULT LEVEL register. After this event has happened, this bit will be reset if the event condition is cleared again. This bit is set when an fb event occurs in even one of the activated channels.

FAULT LEVEL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x0	x	DEV_MAX_CURR_LEVEL [2:0]			x	SHORT_LEVEL [2:0]			x	FB_LEVEL [2:0]		

Bits	Fields	Descriptions
10:8	DEV_MAX_CURR_LEVEL	The DEV_MAX_CURR_LEVEL determines the physical maximum current level of the device. The max dimming sink-current set by the MAX_CURR_VREF can not be beyond the DEV_MAX_CURR_LEVEL. 3b'000 : max 4mA 3b'100 : max 24mA 3b'001 : max 8mA 3b'101 : max 32mA 3b'010 : max 12mA 3b'110 : max 46mA 3b'011 : max 16mA 3b'110 : max 64mA
6:4	SHORT_LEVEL	The SHORT_LEVEL determines the level at which a Short event occurs. 3b'000 : above 3V 3b'100 : above 13V 3b'001 : above 4V 3b'101 : above 16V 3b'010 : above 6V 3b'110 : above 24V 3b'011 : above 8V 3b'111 : above 36V
2:0	FB_LEVEL	The FB_LEVEL determines the level at which a FB event occurs. 3b'000 : above 0.40V 3b'100 : above 0.85V 3b'001 : above 0.50V 3b'101 : above 1.00V 3b'010 : above 0.60V 3b'110 : above 1.15V 3b'011 : above 0.70V 3b'111 : above 1.30V

FAULT MODE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x80F	TIMEOUT_EN	x	x	x	x	x	x	S2_EN	S1_EN	T_OFF_EN	S_OFF_EN	O_OFF_EN

Bits	Fields		Descriptions
11	TIMEOUT_EN	0	The timeout function is disabled. So, if the device will not receive any response from other daisied devices on any read-back commands execution, internal communication logics can not recover its function to the standby state. Recommend this bit to be set.
		1	Internal timeout timer start counting from completion of some read-back commands and stops after receiving the first response bit from daisied devices. Even if the timer will be expired due to any no response, internal communication logics will be recovered to standby state in order to be possible to communicate again. The timeout will be expired at the time of MCLK * 8,192.
4	S2_EN	0	The LED short detection is disabled.
		1	The LED short detection can work under only the LD data’s MSB 5 bits are not all zeros.
3	S1_EN	0	The LED short detection is disabled.
		1	The LED short detection can work under only the LD data’s MSB 4 bits are not all zeros. In case that MSB 4 bits are not all zeros and the SHORT_LEVEL is set to 3V, the short event will be detected if the corresponding channel’s headroom goes above 3V.
2	T_OFF_EN	0	Even if the thermal event happens, all channels’ output will be not off, but the flag can be visible through the FAULT STATUS.
		1	If the thermal event happens, all channels’ output will be off, and the flag can be visible through the FAULT STATUS. For the operating condition, refer to the Electrical characteristics.
1	S_OFF_EN	0	Even if the LED short event happens, the corresponding channel output will be not off, but the flag can be visible through the FAULT STATUS.
		1	If the LED short event happens, the corresponding channel output will be off, and the flag can be visible through the FAULT STATUS. The LED short detection condition can be configurable through the SHORT_LEVEL, the S1_EN bit and S2_EN bit is set.
0	O_OFF_EN	0	Even if the LED open event happens, the corresponding channel output will be not off, but the flag can be visible through the FAULT STATUS.
		1	If the LED open event happens, the corresponding channel output will be off, and the flag can be visible through the FAULT STATUS. The LED open event can be detected if the channel headroom goes down below 0.25V.

DELAY_CH1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x0	x	x	x	x	DELAY_CH1 [7:0]							

DELAY_CH2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0F	0x0	x	x	x	x	DELAY_CH2 [7:0]							

DELAY_CH3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x10	0x0	x	x	x	x	DELAY_CH3 [7:0]							

DELAY_CH4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x11	0x0	x	x	x	x	DELAY_CH4 [7:0]							

DELAY_CH5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x12	0x0	x	x	x	x	DELAY_CH5 [7:0]							

DELAY_CH6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x13	0x0	x	x	x	x	DELAY_CH6 [7:0]							

DELAY_CH7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x14	0x0	x	x	x	x	DELAY_CH7 [7:0]							

DELAY_CH8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x15	0x0	x	x	x	x	DELAY_CH8 [7:0]							

DELAY_CH9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x16	0x0	x	x	x	x	DELAY_CH9 [7:0]							

DELAY_CH10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x17	0x0	x	x	x	x	DELAY_CH10 [7:0]							

DELAY_CH11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x18	0x0	x	x	x	x	DELAY_CH11 [7:0]							

DELAY_CH12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x19	0x0	x	x	x	x	DELAY_CH12 [7:0]							

Bits	Fields	Descriptions
7:0	DELAY_CHx	The DELAY_CHx determines from which sub-frame to start dimming out. For more details, refer to the Figure. 3/4/5.

SERIAL CLOCK GEN

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x408	SERIAL_CLK_LOW [3:0]				x	x	x	SERIAL_CLK_HIGH [4:0]				

Bits	Fields	Descriptions
11:8	SERIAL_CLK_LOW	These bits determine the number of MCLK of V_{DD} level in logic low value ‘0’ or the number of MCLK of zero level in logic high value ‘1’ when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Figure. 1.
4:0	SERIAL_CLK_HIGH	These bits determine the number of MCLK of V_{DD} level in logic high value ‘1’ or the number of MCLK of zero level in logic low value ‘0’ when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Figure. 1.

LD CONTROL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x27	0x848	OFS_TEMP [3:0]				x	ICTL_GLB [2:0]			BGR_CTL [3:0]			

Bits	Fields	Descriptions
		It is recommended to use the default value

DCLK PERIOD

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x28	0x100	x	x	DCLK_PERIOD [9:0]									

Bits	Fields	Descriptions
		The integer multiple of DCLK_PERIOD must be the SF_PERIOD value.

These registers can not go outside the Global Technologies.

OP MODE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3F	0x0	TEST_EN	x	x	x	LD_FIX_EN	x	x	x	x	x	x	ADDR_EXT

Bits	Fields		Descriptions
11	TEST_EN	0	Normal operation
		1	There are two purposes to use this pin as follows : 1. The MCLK divided by 8 is output to the D_{DIO} pin for trimming the internal OSC. 2. The analog value of the MAX_CURR_VREF is output to the pin 17 for trimming. 3. The LD_FIX value is output to internal current control logics without sync signals.
7	LD_FIX_EN	0	Normal operation
		1	There are two purposes to use this pin as follows : 1. The MAX_CURR_VREF reg value is output continuously. 2. The LD_FIX value is output to internal current control logics without sync signals.
0	ADDR_EXT		This bit is used to select which register group, the general registers or the OTP mirror registers, will be accessed. The OTP control registers address, 0x3A ~ 0x3F, can be commonly accessed regardless of this bit.
		0	The general registers can be accessed except for the OTP mirror registers.
		1	The OTP mirror registers can be accessed except for the general registers.

OTP ACCESS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3A	0x0	x	x	x	x	x	x	x	x	OTP_PG_ACCESS_CYCLE [15:12]			

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3B	0x140	OTP_PG_ACCESS_CYCLE [11:0]											

OTP WRITE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3C	0x04	x	x	x	x	x	x	x	x	OTP_WSEL [3:0]			

OTP RD/PROG

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3D	0x0	x	x	x	x	x	x	x	x	x	x	OTP_RD_START	OTP_PG_START

OTP PROTECT

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3E	0x5A5	PROTECT_EN/DIS [11:0] ※ enable (0x5A5), disable (0xA5A), need to disable for writing											

These registers can not go outside the Global Technologies.

OTP CRC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x00	x	x	x	x	OTP_CRC_CHECKSUM [7:0]							

OSC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x40	x	x	x	x	x	OSC [6:0]						

VREF CTL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x10	x	x	x	x	x	x	x	VREF_CTL [4:0]				

OFS 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0x40	x	x	x	x	x	OFS_CH1 [6:0]						

OFS 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x04	0x40	x	x	x	x	x	OFS_CH2 [6:0]						

OFS 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x05	0x40	x	x	x	x	x	OFS_CH3 [6:0]						

OFS 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x40	x	x	x	x	x	OFS_CH4 [6:0]						

OFS 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0x40	x	x	x	x	x	OFS_CH5 [6:0]						

OFS 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x40	x	x	x	x	x	OFS_CH6 [6:0]						

OFS 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x40	x	x	x	x	x	OFS_CH7 [6:0]						

OFS 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0A	0x40	x	x	x	x	x	OFS_CH8 [6:0]						

OFS 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0B	0x40	x	x	x	x	x	OFS_CH9 [6:0]						

OFS 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x40	x	x	x	x	x	OFS_CH10 [6:0]						

OFS 11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x40	x	x	x	x	x	OFS_CH11 [6:0]						

These registers can not go outside the Global Technologies.

OFS 12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x40	x	x	x	x	x	OFS_CH12 [6:0]						

GAIN 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1B	0x40	x	x	x	x	x	GAIN_CH1 [6:0]						

GAIN 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1C	0x40	x	x	x	x	x	GAIN_CH2 [6:0]						

GAIN 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1D	0x40	x	x	x	x	x	GAIN_CH3 [6:0]						

GAIN 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1E	0x40	x	x	x	x	x	GAIN_CH4 [6:0]						

GAIN 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1F	0x40	x	x	x	x	x	GAIN_CH5 [6:0]						

GAIN 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x20	0x40	x	x	x	x	x	GAIN_CH6 [6:0]						

GAIN 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x21	0x40	x	x	x	x	x	GAIN_CH7 [6:0]						

GAIN 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x22	0x40	x	x	x	x	x	GAIN_CH8 [6:0]						

GAIN 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x23	0x40	x	x	x	x	x	GAIN_CH9 [6:0]						

GAIN 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x24	0x40	x	x	x	x	x	GAIN_CH10 [6:0]						

GAIN 11

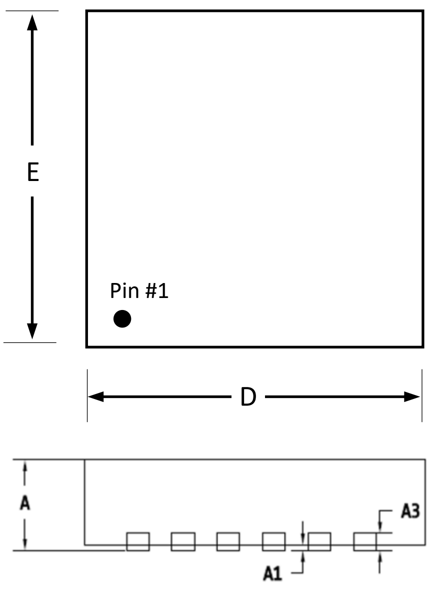
Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x25	0x40	x	x	x	x	x	GAIN_CH11 [6:0]						

GAIN 12

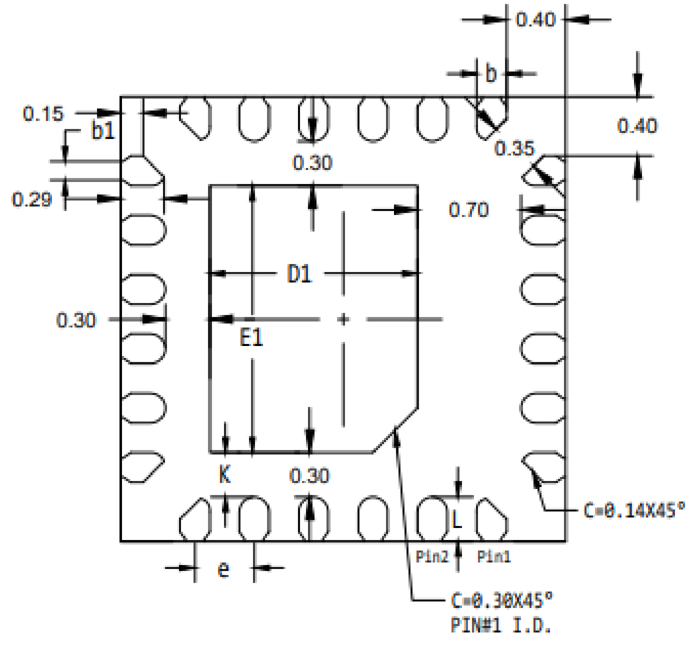
Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x40	x	x	x	x	x	GAIN_CH12 [6:0]						

Package specification

TOP VIEW

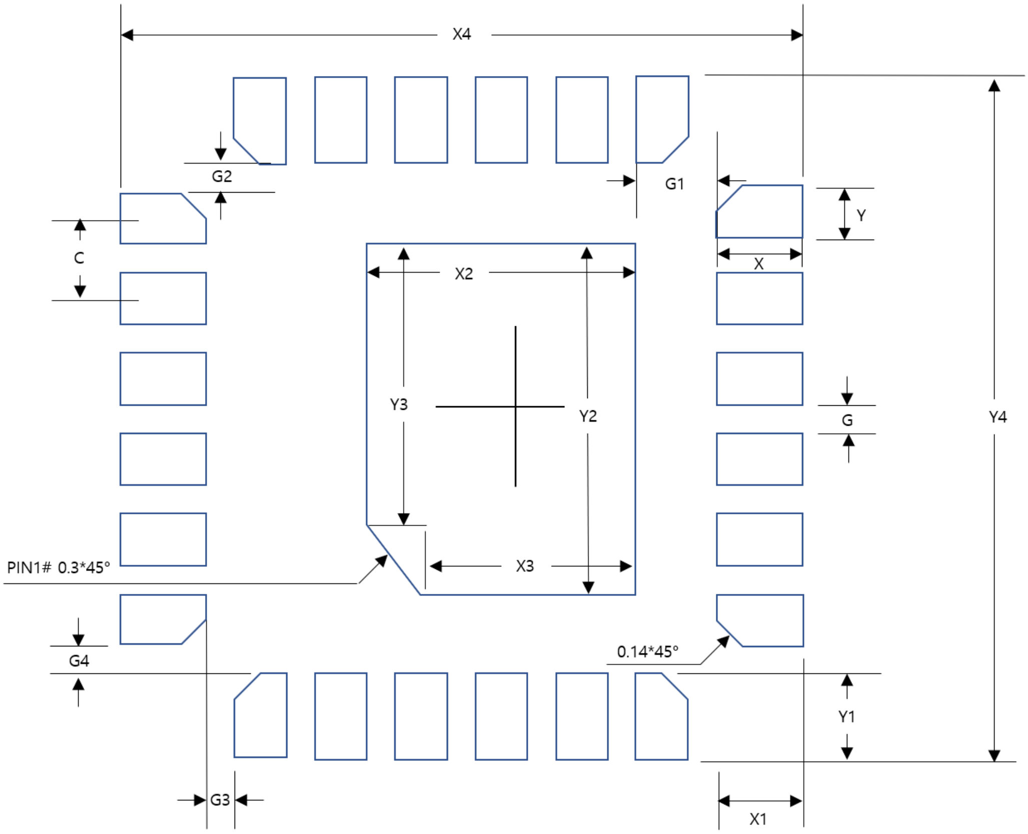


BOTTOM VIEW



Symbol	Dimensions in mm		
	Min	Nom	Max
A	0.50	0.55	0.60
A1	0.00	-	0.05
A3	0.152REF		
b	0.15	0.20	0.25
b1	0.08	0.13	0.18
D	2.90	3.00	3.10
D1	1.25	1.30	1.35
E	2.90	3.00	3.10
E1	1.75	1.80	1.85
e	0.40BSC		
K	0.25	0.30	0.35
L	0.25	0.30	0.35

Footprint



Dim	Value(mm)	Margin(Typ+)
X	0.5	+0.2
Y	0.2	0
C	0.4	0
X1	0.5	+0.2
Y1	0.5	+0.2
X2	1.4	+0.1
Y2	1.9	+0.1
X3	1.1	+0.1
Y3	1.6	+0.1
X4	3.2	+0.2
Y4	3.2	+0.2
G	0.2	0
G1	0.3	0
G2	0.1	0
G3	0.1	0
G4	0.1	0